PROPOSAL: DESIGN AND DEVELOPMENT OF A WEB-BASED INVENTORY AND SALES MANAGEMENT SYSTEM FOR ELKIM CHEMISTRY

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1 INTRODUCTION

1.1 Background

Globally, the use of Information and Communication Technology (ICT) has greatly impacted business operations in various industries that require high accuracy, accountability, and timely decision-making processes, like the pharmaceutical industry (Ma et al., 2022). For instance, modern inventory and sales management systems have enabled various organizations to keep track of their stock in real-time, record sales, generate reports, and reduce losses arising from human error and fraud. Research shows that inaccurate inventory data, stock shortages, and insufficient real-time visibility are associated problems caused by manual inventory systems, which lead to poor operational efficiency and profits. Therefore, many SMEs across the globe are now widely utilizing a centralized, web-based inventory system for the purpose of integrating their operations, increasing transparency.

However, in many developing countries, Kenya included, a good number of small and medium pharmaceutical enterprises still depend on manual methods of inventory management, such as keeping handwritten records and spreadsheets. Despite having spreadsheets as an effective method that provides basic record-keeping functionality, they do not hold a feature for automation, real-time synchronization across multiple branches operating as one enterprise, or secure role-based access control where, an employee is entitled to access their branch's data only and not for all branches, unless authorized, and that the admin is able to view the data for all branches in one platform. The absence of these important dimensions within many Kenyan SMEs, and more so pharmaceutical outlets, exposes them to errors, manipulation of prices, reporting, and poor performance monitoring.

In densely populated areas of Kenya, such as urban settings, for example, Embakasi South and Embakasi East constituencies in Nairobi County, many small pharmaceutical outlets operate in conventional methods, perhaps without fully automated systems for inventory and sales management. In this context, the conventional methods used, such as manual recording on paper files and on spreadsheets, heighten the chances for pricing inconsistencies, where an employee may sell drugs at increased prices to customers who are unaware, unrecorded sales, stock discrepancies, and delayed identification of low-stock items. Since many outlets are managed by their owners, many times, the issue of low stock is very consistent because even if the employees notice it takes a lot of time as they communicate with the owner to arrange for restocking, which consumes a lot of time. Additionally, the business owners struggle to monitor performance by each branch due to the lack of a centralized reporting mechanism, where they rely on the information they receive from their employees when they make a visit to each branch or communicate through phones. This is complex because the owners do not have sufficient time to visit each branch regularly.

From this viewpoint, Elkim Chemistry, a pharmaceutical business outlet owned by Mr. John (fictitious name due to privacy) operates five branches within Nairobi County, with the main branch situated in Embakasi Pipeline, and other two branches also operating in Pipeline but on different areas of Pipeline, while the fourth branch operates in Mukuru Kwa Njenga area, with the fifth branch operating in Tassia in Embakasi East constituency. Elkim encounters operational challenges close to the ones underlined above, whereby it maintains its inventory and sales records through manual methods involving file records and spreadsheets for calculations. Thus, the outlet lacks a centralized system that integrates operations across all branches, such that John can perform business evaluations and track sales and performance for each branch. As

a result of this gap, Elkim faces multiple consequences, including a series of inaccuracies based on stock monitoring, auditing, branch performance comparison, and decision-making, which are time-consuming. In addition, the lack of automation facilitates the absence of low-stock alerts, which can show the drugs that are nearly running out of stock, and this results in unexpected stockouts, forcing loyal customers and one-time customers to seek services elsewhere.

1.2 Introduction: Research Area

The study can be classified as an Information Systems and Software Engineering discipline, with more specific areas of interest being web-based enterprise systems and inventory management solutions. Information systems for enterprises have been developed to integrate business processes into one system for the purpose of real-time monitoring and control, as well as enhanced decision-making processes (Nisrina et al., 2024; Killedar et al., 2024).

Across the globe, there is a shift from separate and manual systems towards centralized web-based systems that offer dashboards, reporting, performance analytics, and user management. These systems help increase transparency, reduce fraud, and increase accountability. In the pharmaceutical retail sector, digital inventory management is of critical importance for ensuring availability, avoiding expiration of medicines, and for effective management of stocks.

Despite these developments, many SMEs in Kenya, especially small pharmaceutical retailers, have not fully embraced and implemented automated enterprise systems. This research, therefore, seeks to bridge the gap between global ICT developments and their actual application in local multi-branch pharmaceutical enterprises. This research adds to the increasing pool of knowledge on how ICT applications can facilitate business operations and governance for small and medium businesses.

1.3 Problem Statement

Elkim Chemistry and similar SMEs, even those operating in other sectors such as retail and hotels in Embakasi South and Embakasi East constituencies of Nairobi county, which form the most densely populated areas in Nairobi, face significant operational inefficiencies due to the reliance on manual methods of inventory and sales management, perhaps including paper or file work and spreadsheets, record keeping, and calculations. The outlet lacks a centralized framework for managing its five branches operating in different locations within the two sub-counties, and this has created accountability gaps, data inconsistencies, and challenges associated with performance monitoring.

The dimensions involving manual recording of sales and inventory intensify the potential for human error by employees working in these branches, whereby they often forget to record some daily transactions, enter incorrect quantities, or often miscalculate the total daily or weekly sales. Since all the branches operate 24 hours a day, with day and night shifts, employees on each shift often blame each other for any issues emerging due to a lack of monitoring on when an issue occurred and who was involved. These errors affect the stock records directly, as well as impacting the generation and storage of clean financial reports and records. Additionally, these manual systems facilitate unethical contact by the staff since an employee may sell drugs at inflated prices to gain personal benefit. On some occasions, employees may plan to avoid recording the sales in a file or spreadsheet and retain the cash, and this may affect others who may be scrutinized to explain the whereabouts of the cash sales and even pay for it when noticed. Since Elkim lacks a centralized automated system to track sales patterns and data in real time, such discrepancies remain undetected until a physical stock count is conducted.

Additionally, stock mismanagement presents a serious issue in all branches of Elkim. On many occasions, drugs are sold without proper documentation, and this results in unexplained shortages. In cases when these issues are noticed, employees are often unfairly accused or penalized for loss of cash sales that have been stolen by their colleagues. Thus, this affects staff morale (among those accused unfairly) and trust between management and staff. Basically, a lack of transparency can facilitate the development of internal conflicts and, hence, diminish overall productivity (Almahraj & Hejazi, 2025). The lack of a system also leads to the absence of performance evaluations for each branch, which can hinder effective decision-making. Indeed, the daily sales cannot be compared in detail, as only the total daily sales from each branch are submitted to the owner in raw figures after each shift. This is therefore linked to limiting data-driven decision-making about timely procurement, pricing adjustments, and branch supervision. Additionally, the staff struggles to identify which drugs are nearing expiry dates and plan for their use in a timely manner to avoid having them expire and get disposed of.

Auditing processes are also complex and time-consuming. Determining the number of drugs under each category, verifying sales totals, and reconciling stock discrepancies require manual effort. The absence of automated low-stock alerts means that essential drugs may run out unexpectedly. When customers find medicines unavailable, they often seek alternatives from competing pharmacies, leading to loss of revenue and potential long-term customers. The magnitude of these issues impacts financial performance, customer satisfaction, employee relations, and the sustainability of the business. There, therefore, exists a need for a central, automated system to improve transparency and accountability, as well as operational efficiency in all branches of the business.

1.4 Proposed Solution

The research project seeks to design and develop a centralized web-based inventory and sales management system for Elkim Chemistry that will serve to automate all the necessary inventory dimensions, including inventory tracking, sales recording, payment monitoring, and conducting evaluations for branch performance. The solution is aimed at replacing the manual file/paper and spreadsheet-based record keeping with an integrated platform accessible to all branches through a secure role-based access control. The system will function to centralize Elkim's operations across all its five branches into a unified database interface, allowing the admin (the business owner) to monitor stock levels, sales performance, payment modes (either cash or M-PESA), drugs nearing out of stock, drugs nearing expiry dates, and all other branch activities in real time through a dashboard interface.

The components of research specific to this project will involve analyzing the best practices in modern inventory management systems, pinpointing the functional requirements unique to Elkim Chemistry, and evaluating the effectiveness of the developed system in enhancing operational efficiency. The solution will integrate automated stock updates after each sale, low-stock alert mechanisms as per the predefined thresholds, expiry dates' alert mechanisms based on the predefined dates for alerts, and branch-level performance reporting. By aligning with global enterprise system models that emphasize transparency, centralization, and data-driven decision-making, this research will demonstrate how SMEs in Nairobi can leverage ICT solutions to overcome operational inefficiencies and strengthen internal controls.

1.5 Objectives

General Objective: To design, develop, and evaluate a centralized web-based inventory and sales management system that enhances efficiency, transparency, and branch coordination at Elkim Chemistry within a one-and-a-half-month period.

Specific Objectives

1. To analyze the existing inventory and sales management processes at Elkim Chemistry and identify operational weaknesses within the first week.
2. To design a centralized system architecture integrating all five branches with role-based access control by the second week.
3. To develop and implement a web-based inventory and sales management system that automates stock updates, sales recording, and payment tracking by the fourth week.
4. To incorporate a low-stock alert mechanism that notifies management when drug quantities fall below predefined thresholds and an alert mechanism that notifies when a drug is nearing expiry date.
5. To evaluate the system's effectiveness in reducing discrepancies, improving reporting accuracy, and enhancing branch performance monitoring by the eighth month.
6. To document findings and recommend best practices for similar SMEs seeking digital transformation.

1.6 Justification

The importance of this research is seen in relation to Elkim Chemistry, other SMEs in Nairobi, and Information Systems in general. To Elkim Chemistry, the system is important because it will help reduce inaccuracies in inventory and sales records, minimize losses from unrecorded transactions, and increase accountability. The centralized dashboard will enable the

owner to monitor branch performance in real time, detect irregularities early, and make informed procurement and pricing decisions. Other small and medium enterprises that use similar manual solutions can look to this study to digitize their entire operations. This study will show how using a web-based solution can increase transparency, improve customer satisfaction, and enhance operational efficiency for any enterprise facing limited resources.

Academically, the research contributes to the application of software engineering principles in solving real-world business problems. It bridges theoretical ICT knowledge with practical implementation in a local context.

1.7 Proposed Research & System Methodologies

This study will be based on a structured methodology that will involve a sequence of phases starting with gathering the requirements, performing comprehensive system analysis and physical design, developing the system and implementing it, testing the developed system, and evaluating the system functionality, which will be based on determining if the system has addressed all the user requirements. In this context, data that will lead to gathering the requirements will be collected through conducting an interview with the business owner and the staff, directly observing the existing processes, and analyzing the current manual and spreadsheet records systems. In the system development phase, the Software Development Lifecycle (SDLC) iterative model will be employed. This model will allow for progressive refinement of system components based on the feedback. The system modeling will be informed by the principles of Object-Oriented Analysis and Design (OOAD), which will guarantee modularity, scalability, and the maintainability of the system. These methodology approaches will be appropriate because they will provide systematic procedures for translating the user requirements into a reliable and functional system.

1.8 Scope

Geographically, this study will focus on the five branches of Elkim Chemistry located in Embakasi South and Embakasi East constituencies within Nairobi County. The target users include the business owner (Administrator) and branch staff responsible for inventory and sales operations. Functionally, the system will cover inventory management, sales recording, payment mode tracking (Cash and M-PESA), low-stock alerts, expiry dates alerts, and branch performance reporting. The study will not include patient medical records, prescription management, supplier integration systems, or clinical decision-support tools. It will strictly focus on inventory and sales management operations within the selected enterprise.

1.9 Project Time Plan

The project will run from *5th February to 20th March*, excluding weekends. The development will follow a structured sequence beginning with requirements gathering and ending with system implementation and documentation. See *Appendix A*, representing the Gantt Chart with specific durations for each task.

1.10 Resources

Appendix B shows the resources to be utilized in this project and how each resource will be important and utilized in the project.

1.11 Budget and Justification

The proposed budget is strictly on the technical requirements that are essential for the successful development, deployment, and safe operation of the web-based system. The purchase of a 32GB flash disk is essential for the safe deployment of the system source code, database files, and project documentation, thereby reducing the risk of data loss during the development process. The registration of the domain name for one year will enable the deployment of the

system on an accessible web platform, thereby allowing remote access for the supervisor and the business owner. The need to upgrade the RAM of the laptop from 8 to 16 GB arises in order to improve its performance, since development tools like Visual Studio that will provide an Integrated Development Environment (IDE) for coding, and SQL Server consume a lot of memory resources when running simultaneously. A contingency allocation will be included to cater to any minor technical expenses that might arise unexpectedly during the configuration and implementation of the project. Since internet connectivity, transport, and printing services are already available, they will not be included in the budget. See *Appendix C* for the proposed budget.

1.12 Project Monitoring and Evaluation

The monitoring of the project will be done on a continuous basis throughout the development process to ensure that all tasks are completed within the stipulated period and in line with the objectives. This can be achieved through regular checks against the Gantt chart to ensure that all phases of the project, from requirement gathering to implementation, are completed within the stipulated period. Additionally, regular consultations will be done with the project supervisor to seek guidance and ensure that all tasks completed are in line with the objectives and meet the academic standards. Prior to commencing each phase of the project, completed tasks will be checked to ensure that they meet the stipulated standards and expectations. Testing checks will also be done after completing each module of the project. Project evaluation will focus on assessing whether the developed system effectively addresses the identified problems. Key evaluation criteria will include the system's functionality in meeting user requirements, the accuracy of inventory tracking, the reliability of automated low-stock alerts, and the usability of the system for both the administrator and branch staff. The

ability of the system to generate accurate sales and stock reports, as well as to provide centralized monitoring of branch operations, will also be assessed. Final evaluation will involve a practical demonstration of the system to the business owner, followed by documentation of feedback and recommendations for further improvement.

2 References

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3 Appendices

3.1 Appendix A: Project Time Plan: *Gantt Chart*

Task/Duration	Feb 5-Feb 8	Feb 9-Feb 14	Feb 15-Feb 21	Feb 22-Feb 26	Feb 27-Mar 11	Mar 12-Mar 15	Mar 16-Mar 17	Mar 18-Mar 19
Requirements Gathering								
System Analysis								
System Design								
Interface Design								
System Development								
System Testing								
System Implementation								
Documentation and Final Review								

3.2 Appendix B: Resources

Human Resource	Role/Responsibility
Project Developer (Student)	Responsible for system analysis, design, development, testing, implementation, and documentation.
Business Owner (Administrator)	Provides system requirements, validates functionality, and offers feedback during testing and evaluation.
Branch Staff	Assist in clarifying operational procedures and participate in user testing of the system.
Hardware Resource	**Purpose**
Personal Laptop/Computer	Used for system design, development, testing, and documentation.
Internet Connectivity	Required for research, deployment, updates, and remote system access.
External Storage Device (Flash Disk)	Used for secure backup of source code, database files, and documentation.
Software Resource	**Purpose**
Sublime Text	Development environment for building the web-based system.
SQL Server	A database management system for storing inventory and sales data.
Microsoft Office	Preparation of project documentation and reports.
Web Browser	Testing and demonstration of the web-based system.

3.3 Appendix C: Proposed Budget

Item	Estimated Cost (Ksh)
32GB Flash Disk (External Backup)	1,500
Domain Name Registration (1 year)	1,500
RAM Upgrade (8GB to 16GB)	4,500
Contingency (Unforeseen technical expenses)	1,500
Total Estimated Budget	**9,000 Ksh**